

MATH 351 final exam checklist

For the final exam, you bring three sides of handwritten notes. You may use a calculator of your choice, as on the first and second midterm exams. I will provide a standard normal table with the exam.

General

- I have read all the assigned sections of the textbook.
- I have read all the feedback on Blackboard for my homework assignments.
- I have studied all the solution sets for homework assignments on Blackboard.
- I have studied all the solution sets for quizzes on Blackboard.
- I have studied the solution sets for the two midterm exams on Blackboard and understand any questions where I missed points on those exams.
- On my sheet of notes, I have written information about the following standard distributions, including common situations where each distribution comes up, possible values, probability formulas (for each value in discrete case; density and distribution functions in continuous case), and formulas for expected value and variance.
 - binomial(n, p)
 - Bernoulli(p)
 - geometric(p)
 - Poisson(λ)
 - Urn-problem distribution – Durrett calculates the variance, but we did not talk about it.
 - Uniform distribution on an interval $[a, b]$
 - exponential(λ)
 - normal(μ, σ^2)

Formalism of probability

- I know the formalism for describing chance experiments: outcomes, sample space, events.
- I know the operations of the algebra of events and can use these operations to express complex events in terms of simpler events. For example, the event “at least one of A, B, C occurs” means “ $A \cup B \cup C$ ”.
- I know that we assign probabilities to events.
- I know the rules of probability, including the basic rules and derived rules such as $P(A^c) = 1 - P(A)$ and $P(A \cup B) = P(A) + P(B) - P(A \cap B)$.
- I know the formal definition of “A and B are independent.”
- I know what “A and B are independent” means informally.
- I can identify independence assumptions when making probability models of real-world situations and explain how I use those assumptions in my calculations.
- I know when one adds probabilities of events and when one multiplies probabilities of events.

Random variables

- I know the formal definition of a random variable.
- I know the formal definition of the distribution of a random variable.
- I can calculate the distribution of a random variable given a description chance experiment and a random variable for that experiment.
- I can use the “ $\sim$ ” notation to indicate that a random variable has one of our named distributions (binomial, Poisson, geometric so far).
- I know the formal definition of the expected value of a random variable.
- I know how to interpret expected value informally, as
- I can calculate the expected value of a random variable given its distribution.
- I know what the moments of a random variable are, and I can calculate them given the distribution of a random variable using the Law of the Unconscious Statistician (LOTUS).
- I know how to calculate the variance of a random variable X using $E[X^2]$ and $E[X]$.
- I know how to calculate $\sigma(X)$ using $\text{Var}(X)$ for a random variable X .
- I know how to match a random variable described in terms of a chance experiment with the story of the binomial distribution.
- I know the formula for the binomial-distribution probabilities.
- I know the Poisson approximation formula for $\text{binomial}(n, p)$ where n is large and p is small.
- I know the $\text{Poisson}(\lambda)$ distribution.
- I can use the Poisson distributions model counting rare events.
- I know the formula for the $\text{geometric}(p)$ distribution.
- I know how to match the story of a “waiting time” random variable with the $\text{geometric}(p)$ distribution.
- I know the shortcut formulas for expected value and variance of random variables with the $\text{binomial}(n, p)$, $\text{Poisson}(\lambda)$, and $\text{geometric}(p)$ distributions.
- I can solve “urn problems,” matching chance experiments with the story of drawing balls from an urn with a certain number of red balls, using the random variable that counts the number of red balls drawn.

Counting

- I can count using the multiplication rule when counting done in stages with “homogeneity” in the possibility tree at each stage.
- I can count by matching a counting situation with the stories of permutations and combinations.
- I can use Durrett’s terminology and notation for permutations and combinations.
- I can count using the multinomial counting method.
- I can combine counting methods to solve complex problems, such as splitting a class into groups where the order does not matter or making license plates where letters and numbers do not have fixed positions.
- I can use these counting methods to calculate probabilities in situations where all outcomes are equally likely.

Conditional probability

- I know the definition of conditional probability: for events A and B where $P(B) > 0$, the probability of A given B is $P(A|B) = P(A \cap B) / P(B)$.
- I know that to say events A and B are independent (where $P(B) > 0$) is equivalent to saying $P(A | B) = P(A)$.
- I can calculate conditional probabilities using the definition.
- I can calculate conditional probabilities using the “reducing the sample space” method.
- I can reverse the definition of conditional probability and use it as a multiplication rule for calculating $P(A \cap B)$ when A and B are not necessarily independent: $P(A \cap B) = P(A) P(B|A)$.
- I know that a partition of a sample space is a list of events $B_1, \dots, B_n$ that are pairwise disjoint (for any i and j with $i \neq j$, $B_i \cap B_j = \emptyset$) and whose union is the whole sample space ($B_1 \cup \dots \cup B_n = \Omega$).
- I know that for any event B, the events B and B^c form a partition of the sample space.
- I can use the law of total probability to calculate $P(A)$ given a partition $B_1, \dots, B_n$ of a sample space: $P(A) = P(B_1) P(A|B_1) + \dots + P(B_n) P(A|B_n)$. I know that with this problem-solving technique, we think of our chance experiment as a “two-stage experiment.”
- I know sophisticated examples of the two-stage-experiment technique including cases where A is related to a random variable X and we know the distribution of X conditional on the occurrence of each event B_i in our partition. An example is the random variable X where if event B_i occurs, we flip i coins and count the number X of heads. In this example, we know the distribution of X conditional on B_i ($X|B_i \sim \text{binomial}(i, 1/2)$) and can use the law of total probability to calculate unconditional probabilities related to X.
- I know the Bayes’-formula method for calculating conditional probabilities. I can explain the method as in Durrett, Quiz 7, and various homework questions.
- I know what the joint distribution of two discrete random variables is. I can calculate the joint distribution and present it in a table as in Durrett and homework.
- I can calculate the marginal distributions (separate distributions for X and Y) given the joint distribution of discrete random variables X and Y.
- I know what it means for two random variables to be independent. I can decide whether two discrete random variables are independent using their joint distribution.
- I can use the joint distribution of discrete random variables X and Y to calculate conditional probabilities such as $P(X = i | Y = j)$ and conditional expectations such as $E[X | Y = j]$.

Continuous random variables

- I know the informal definition of a continuous random variable as a random variable that can take any value on the real number line or perhaps any value in an interval of the number line.
- I know that for a continuous random variable X, we consider events of the form $(a \leq X \leq b)$.
- I know that a continuous random variable X has an associated density function f such that
$$P(a \leq X \leq b) = \int_a^b f(x)dx.$$
- I know that a density function f must satisfy $\int_{-\infty}^{\infty} f(x)dx = 1$ (improper integral) and that f must have non-negative values.

- I know that for any random variable X , the distribution function F for X is defined by $F(x) = P(X \leq x)$. I know that distribution functions have limit 0 as $x \rightarrow -\infty$, have limit 1 as $x \rightarrow +\infty$, and are non-decreasing.
- I know that for a continuous random variable with density function f , $F(x) = \int_{-\infty}^x f(x)dx$ (improper integral). I know that in this case, the distribution function F is continuous and $F' = f$. (To make that statement precise requires care, for example, at points where F is not differentiable; we will not consider those details.)
- I know that for a continuous random variable X , $P(X \leq x) = P(X < x)$.
- I know how to calculate $E[g(X)]$, for a continuous random variable X with density function f . (LOTUS for continuous random variables.) In particular, I know how to calculate $E[X]$, $E[X^2]$, and $\text{Var}(X)$ using integrals involving f .
- For independent random variables X and Y , I know that $P(\max(X, Y) \leq x) = P(X \leq x) \cdot P(Y \leq x)$. Using this formula, I can calculate probabilities of events such as “the largest rainfall in the rainforest in a week is at most 1 inch.” (These calculations are not just for continuous random variables, but we practiced them in that context.)
- I know how to pick the parameter λ to match an exponential(λ) random variable X to quantitative data by matching $E[X] = \frac{1}{\lambda}$ with the mean of the data values. (This way of choosing parameters in a probability model for data is sometimes called “the method of moments.”)
- I know formulas for the density function, the distribution function, the expected value, and the variance of a random variable that is uniformly distributed on the interval $[a, b]$ (for real numbers a and b).
- I know that a median of a continuous random variable X is a real number m such that $P(X \leq m) = \frac{1}{2}$ or, equivalently, $F(x) = \frac{1}{2}$, where F is the distribution function for F . I can solve the equation $F(x) = \frac{1}{2}$ using algebra or precalculus when F is not too complicated.
- I know formulas for the median of continuous uniform random variables and exponentially distributed random variables.
- I know formulas for the density function, the distribution function, the expected value, and the variance of a random variable with the exponential(λ) distribution.

Sums of random variables and limit theorems

- I know the formulas $E[X + Y] = E[X] + E[Y]$ and $E[aX] = aE[X]$, valid for any random variables X and Y that have expected values.
- I know how to calculate covariance of random variables X and Y as $\text{cov}(X, Y) = E[XY] - E[X]E[Y]$.
- I know the formula $\text{Var}(aX) = a^2\text{Var}(X)$.
- I know the sum formula for variance, $\text{Var}(X + Y) = \text{Var}(X) + \text{Var}(Y) + 2\text{cov}(X, Y)$.
- I know that if X and Y are independent random variables, then $\text{cov}(X, Y) = 0$.
- I know the abbreviation “iid” for “independent and identically distributed.”
- I can use the expected-value and variance formulas to compute expected values and variances for sums and averages of lists of iid random variables. I can justify the steps of these calculations using those formulas.

- I know that the weak law of large numbers says informally that for a sequence $X_1, X_2, \dots$ of iid random variables (with an expected value and variance), if n is very large, then the average $\overline{X_n} = \frac{1}{n}(X_1 + \dots + X_n)$ is very likely to be very close to $E[X]$. I know that the strong law of large numbers says that “with probability 1,” the sequence $X_1, X_2, \dots$ converges to $E[X]$.
- I know the Bernoulli(p) distribution (another name for the binomial(1, p) distribution). If $X \sim \text{Bernoulli}(p)$, then X has values 0 and 1, $P(X = 1) = p$, and $P(X = 0) = 1 - p$.
- I know what the indicator random variable of an event A is.
- I know that the indicator random variable for an event A having $P(A) = p$ has the Bernoulli(p) distribution.
- I can use the method of indicators to compute expected values of a random variable N . I can explain the steps of the method, including identifying relevant events, calculating probabilities of those events, writing N as the sum of indicators for the events, and using additivity of expected value to calculate $E[N]$.
- I know that a sum of n independent Bernoulli(p) random variables $X_1 + \dots + X_n$ is a binomial(n, p) random variable. This fact explains the formulas for expected value and variance of a binomial random variable.
- I know that if X and Y are independent random variables where $X \sim \text{binomial}(n, p)$ and $Y \sim \text{binomial}(m, p)$ (same p !), then $X + Y \sim \text{binomial}(n + m, p)$.
- I know that if X and Y are independent random variables where $X \sim \text{Poisson}(\lambda)$ and $Y \sim \text{Poisson}(\mu)$, then $X + Y \sim \text{Poisson}(\lambda + \mu)$.
- I know what it means for a random variable to have the normal(μ, σ^2) distribution.
- I know that if X and Y are independent random variables where $X \sim \text{normal}(\mu_1, \sigma_1^2)$ and $Y \sim \text{normal}(\mu_2, \sigma_2^2)$, then $X + Y \sim \text{normal}(\mu_1 + \mu_2, \sigma_1^2 + \sigma_2^2)$.
- I can calculate probabilities of events of the forms $P(X \leq b)$, $P(a \leq X)$, and $P(a \leq X \leq b)$, where $X \sim \text{normal}(\mu, \sigma^2)$, using a standard normal table.
- I know the Central Limit Theorem (CLT) in its formal version,

$$\lim_{n \rightarrow \infty} P\left(a \leq \frac{\overline{X_n} - E[X_i]}{\sigma(X_i)/\sqrt{n}} \leq b\right) = P(a \leq \text{standard normal} \leq b),$$

where $X_1, X_2, \dots$ is a sequence of iid random variables.

- I can use the CLT as an informal problem-solving tool, treating a sum of a large number n of iid random variables $X_1 + \dots + X_n$ as normally distributed.
- I can use histogram corrections to improve CLT estimates of probabilities of events related to a discrete random variable X having integral values. The general principle of these corrections is to think of an event $X \leq n$ as $X \leq n + \frac{1}{2}$ and to think of an event $X \geq n$ as $X \geq n - \frac{1}{2}$ (for an integer n).
- I know how to test a probability model for a real-world situation by calculating a p-value. (One can often use the CLT to help calculate the p-value.)